

Q1. (5 marks)

- (a) Let K, L and M be statement variables and suppose that the following logical expression is TRUE.

$$\neg([L \vee (\neg M)] \Rightarrow (\neg K)) \wedge (\neg L)$$

Determine with justification all possible truth value combinations of K, L, M .

- (b) Let G, H, J be statement variables. Use a truth table to prove that

$$[H \wedge (G \Leftrightarrow J)] \quad \equiv \quad [G \wedge (H \wedge J)] \vee \neg[H \Rightarrow (G \vee J)].$$

Q2. (6 marks) Consider the implication

$$J(q, r) : \text{“If } q + 2 \neq 0 \text{ or } r^2 - 1 \neq 0 \text{ then } (q + 2)(r^2 - 1) \neq 0.”}$$

Answer the following questions. **No justification is necessary.**

- (a) State the hypothesis of $J(q, r)$.
 (b) State the conclusion of $J(q, r)$.
 (c) Let $L(q, r)$ be the contrapositive of $J(q, r)$. State $L(q, r)$, and give the truth value of

$$\forall q \in \mathbb{R}, \forall r \in \mathbb{R}, L(q, r).$$

- (d) Let $K(q, r)$ be the converse of $J(q, r)$. State $K(q, r)$, and give the truth value of

$$\forall q \in \mathbb{R}, \forall r \in \mathbb{R}, K(q, r).$$

- (e) Express the following statement symbolically, without using words, the symbol $J(q, r)$ or the $\neg$ symbol.

$$\forall r \in \mathbb{R}, \exists q \in \mathbb{R}, J(q, r)$$

Is this statement true or false?

- (f) State the truth value of

$$\forall q \in \mathbb{R}, \forall r \in \mathbb{R}, \left[\left((q + 2 \neq 0) \wedge (r^2 - 1 \neq 0) \right) \iff \left((q + 2)(r^2 - 1) \neq 0 \right) \right].$$

Q3. (5 marks) For each of the following statements, clearly state whether it is true or false. Then prove or disprove it.

(a) There exist five consecutive integers whose average is equal to the second-largest of those integers.

(b) The average of three consecutive perfect squares is always a non-integer.

Note: A perfect square equals $k \times k$ for some integer k .

Q4. (4 marks) Prove the following statement:

$$\text{For all } x \in \mathbb{R}, 2|5x + 15| - 4|3x - 18| \leq 90.$$

Note: Graphical proofs will not be accepted.

Q5. (5 marks) Prove the following statement

$$\forall p, q, r \in \mathbb{R}, 2(p^2 + 13r^2) + q(5q + 4p - 10r) \geq 2pr.$$

Q6. (5 marks) Fill in each of “open sentence 1”, ..., “open sentence 6”, with an open sentence from the table below, so that the statements written in part (a) and (b) are both true statements.

Each open sentence in the table must be used exactly once.

$M(x) : x^2 + 10x + 24 = 0$	$P(x) : x^2 - 36 = 0$	$Q(x) : x^2 - 10x + 24 = 0$
$S(x) : x^2 - 26 = 10$	$T(x) : x^2 - 16 = 0$	$U(x) : (x - 4)^2(x - 6) \neq 0$

(a) $\forall x \in \mathbb{R}, \left[\left(\frac{\text{open sentence 1}}{\text{open sentence 1}} \wedge \frac{\text{open sentence 2}}{\text{open sentence 2}} \right) \Leftrightarrow \left(\frac{\text{open sentence 3}}{\text{open sentence 3}} \right) \right].$

(b) $\neg \left[\forall x \in \mathbb{R}, \frac{\text{open sentence 4}}{\text{open sentence 4}} \Rightarrow \left(\frac{\text{open sentence 5}}{\text{open sentence 5}} \vee \frac{\text{open sentence 6}}{\text{open sentence 6}} \right) \right].$

(c) Write a statement that is logically equivalent to your statement from (b); it should not include words nor any of the symbols $\neg, \Leftrightarrow, \Rightarrow, \Leftarrow, M, P, Q, S, T, U$.

(d) Prove your statement from (c).